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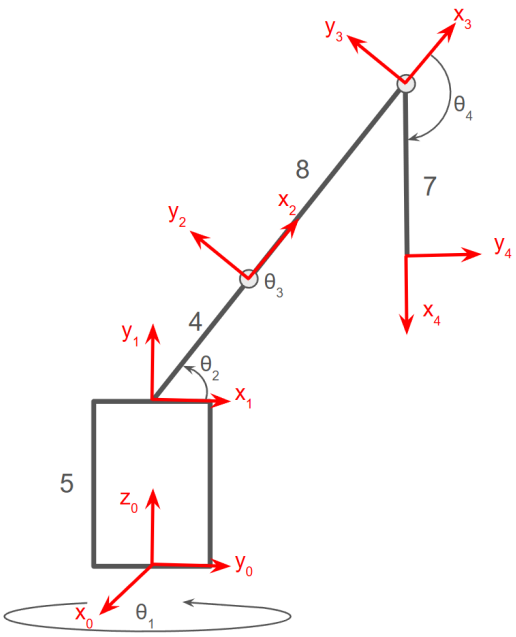
Department of Computer Science and Engineering

CSE 461: Introduction to Robotics

Examination: Quiz II **Duration: 30min**

Semester : Summer 26 **Full Marks: 15**

Name:	ID:	Section:

1	 The manipulator joints rotate 35, 50, 0, and 140 degrees to reach the final point. In this position, the last link is directly perpendicular to the X-axis. A. What do you understand by 3-DOF? **[1]** B. Based on type of movement, how many types of joints are there and what are they? **[2]** C. Form the DH table for the given manipulator **[4]** D. Find the location of the end effector. Clearly mention the x, y, z co-ordinates. **[8]**	**15**

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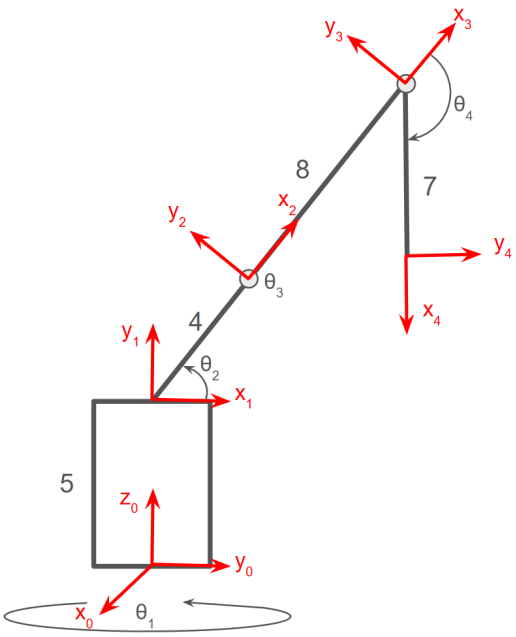
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Examination: Quiz II **Duration: 30min**

Semester : Summer 26 **Full Marks: 15**

Name:	ID:	Section:

1	 The manipulator joints rotate 35, 50, 0, and 140 degrees to reach the final point. In this position, the last link is directly perpendicular to the X-axis. A. What do you understand by 5-DOF? **[1]** B. Based on the type of mobility, how many types of joints are there and what are they? **[2]** C. Form the DH table for the given manipulator. **[4]** D. Find the location of the end effector. Clearly mention the x, y, z co-ordinates. **[8]**	**15**

Rubric

1	Proper explanation	1
2	Number of types correct Correct types mentioned	1 1
3	For each correct parameter	0.25*16 = 4
4	Each matrix correct Each multiplication correct Final location mentioned clearly	1*4 = 4 1*3 = 3 1